EX01 - Dynamic Pressure - First Variables and Decisions

Beginner Step-by-Step Guide

This guide follows the real file `ex01_dynamic_pressure.py`. It explains the problem first and then the code from the perspective of someone starting Python.

1. What we want to solve

Calculate dynamic pressure q when density rho and speed are known.

Problem variables / elements

Name	Meaning
rho	fluid density [kg/m^3]
speed	speed magnitude [m/s]
q	dynamic pressure [Pa]

Equation or logic on paper

```
q = 0.5 * rho * speed**2
```

Manual check

With rho=1.225 and speed=100: speed**2=10000; 0.5*1.225=0.6125; q=6125 Pa.

2. What we already know and what is new

Already seen:

New in this exercise:

- variables and assignment with =
- float
- operators * and **
- comparisons
- if / elif / else
- print() and f-string

3. New Python concepts

1. Variable and assignment

Syntax: `rho = 1.225`

The = sign means assignment: Python evaluates the right-hand side and binds that value to the name on the left. It is not mathematical equality.

```
speed = 100.0
```

2. float number

Syntax: `1.225`

A float is a number with a decimal part. Many engineering quantities are represented this way.

3. Exponentiation

Syntax: `speed**2`

** is exponentiation. speed**2 means speed squared.

4. Conditional

Syntax: `if condition:`

After if we write a condition that evaluates to True or False. The colon opens the block and indentation shows which lines belong to it.

5. f-string

Syntax: `f"{dynamic_pressure} [Pa]"`

The f before the quotes allows values inside braces to be inserted into text.

4. Code step by step

Step 1. Define the inputs

```
rho = 1.225
speed = 100.0
```

We create two names. Unit comments help the reader, but Python does not understand physical units.

What happens internally: Python creates two numeric objects and the names rho and speed become available to later lines.

Step 2. Validate rho

```
if rho < 0:
    print("Please introduce a physical density")
```

Python evaluates rho < 0. With 1.225 the result is False, so the indented block is skipped.

Step 3. Validate speed

```
elif speed < 0:
    print("Please introduce a physical speed")
```

elif is checked only if the previous if was False.

Step 4. Calculate q

```
else:
    dynamic_pressure = 0.5 * rho * speed**2
```

else represents the remaining case: both inputs passed validation. The paper equation translates almost directly into Python.

Step 5. Display the result

```
print(f"Dynamic pressure of the system: {dynamic_pressure} [Pa]")
```

print() sends text to the terminal and the f-string inserts the computed value.

5. Execution flow

```
rho, speed
   -> validate rho
   -> validate speed
   -> calculate q
   -> print
```

6. Expected result / check

With the exercise values the result is 6125.0 Pa.

7. Common beginner mistakes

- Using ^2 instead of **2.
- Forgetting the colon after if/elif/else.
- Not indenting the block.
- Thinking [Pa] or [m/s] are units Python understands.

8. If you remember only five things

- = assigns a value to a name.
- ** performs exponentiation.
- A condition becomes True or False.
- if/elif/else selects one branch.
- The equation can be translated directly into Python operators.

9. Mini-practice

Change one input. Before running, write a comment predicting what should change and why. Then run the file and compare your prediction with the output.

Completion criterion

Move on when you can explain the problem, equation/logic, new Python concepts and execution flow without simply reading the code.